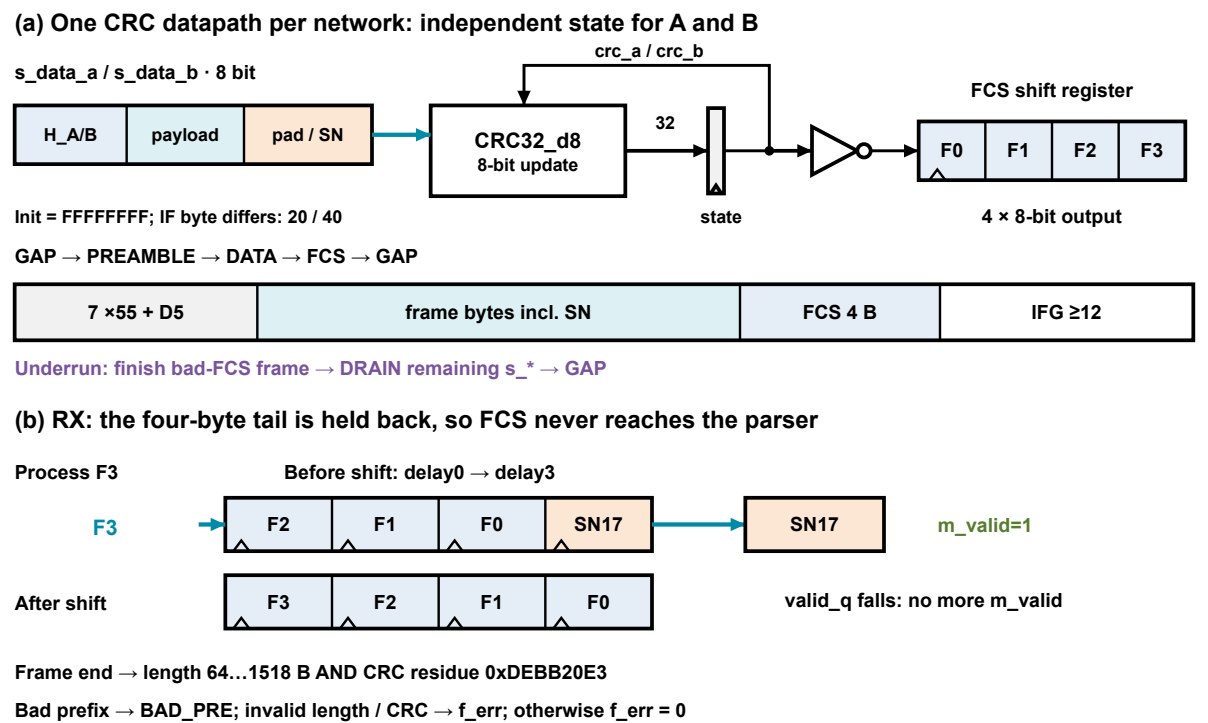
(a) One CRC datapath per network: independent state for A and B
s_data_a / s_data_b · 8 bit
crc_a / crc_b
FCS shift register
H_A/B
payload
pad / SN
CRC32_d8
8-bit update
32
F0
F1
F2
F3
Init = FFFFFFFF; IF byte differs: 20 / 40
state
4 × 8-bit output
GAP → PREAMBLE → DATA → FCS → GAP
7 ×55 + D5
frame bytes incl. SN
FCS 4 B
IFG ≥12
Underrun: finish bad-FCS frame → DRAIN remaining s_* → GAP
(b) RX: the four-byte tail is held back, so FCS never reaches the parser
Process F3
Before shift: delay0 → delay3
F3
F2
F1
F0
SN17
SN17
m_valid=1
After shift
F3
F2
F1
F0
valid_q falls: no more m_valid
Frame end → length 64…1518 B AND CRC residue 0xDEBB20E3
Bad prefix → BAD_PRE; invalid length / CRC → f_err; otherwise f_err = 0